Metadata: WIP aquatic invertebrate resolved-taxa outputs

1. What these files are

Aquatic invertebrate counts from the ABMI / DUC Wetland Invertebrate (WIP) samples, with **ambiguous taxa resolved to a common taxonomic level** so that samples can be compared. Many specimens can only be identified to a coarse level (for example `Chironomidae.UID`, `Corixinae.UID`). The resolution algorithms (Ermias Azeria's `aptc_abmi` and `dpac_abmi`) reassign each such "parent" count to its identified "children" using the taxonomy lookup table.

One workbook = one year x one algorithm run. File names:

```
WIP <year> resolved taxa_<method>_<region>.xlsx
```

Part	Values	Meaning
`year`	2023, 2024, 2025	Sampling year
`method`	`aptc_abmi`	Assign Parents To Children: at a site that has children, the whole parent count goes to the single most numerous child at that site
	`dpac_abmi`	Distribute Parents Among Children: the parent count is split among the children at that site in proportion to their counts
`region`	`frequent`	If a site has no children for a parent, use the child that is most widespread (found in the most sites) across the data (Chao2-like)
	`abundant`	Same fallback, but use the child with the highest total abundance (Chao1-like)

All four combinations are exported because the final choice is still TBD. The 2023 WIP Excel workflow was DPAC-SG; the CABIN pipeline used APTC-frequent.

Fallback regions: every WIP site is treated as Parkland natural region (one stratum). No natural subregions are available, so the fallback uses all sites.

2. Sheets in each workbook

Every workbook has the same 12 sheets. **Rows are samples, columns are taxa**, except `AP.info`. The first column has no header; it holds the sample ID (it was written from R row names).

Sheet	Use it for	Contents
`Out.combined`	**Main analysis table**	Resolved fine data plus resolved UMOS. Equals `Out.main + Out.aux`
`Out.main`		Resolved fine (main) data only, no UMOS
`Out.aux`		Resolved UMOS data only
`Raw.main`	Audit	Input to the algorithm: fine counts after the name crosswalk and scaling, before resolution. Contains both children and ambiguous parents. **Not unscaled raw counts**
`Raw.aux`	Audit	Input UMOS counts (not scaled), same columns as `Raw.main`
`Raw.child`	Audit	Fine counts of taxa that were already resolved (non-parent columns of `Raw.main`)
`New.child`	Audit	Fine counts added to children by redistributing parents. `Raw.child + New.child` plus `Nochild.data` = `Out.main`
`Raw.child.aux`	Audit	Same as `Raw.child`, for UMOS
`New.child.aux`	Audit	Same as `New.child`, for UMOS
`AP.info`	QA	One row per ambiguous parent found in the data: `Parent` and the number of children found for it (`Number of children`; R's `read.xlsx` shows it as `Number.of.children`). Has a blank first column (row

Sheet	Use it for	Contents
		names). It is not samples x taxa
`Nochild.data`	QA	One column per ambiguous parent, filled only for parents with **no children**; fine counts that could not be redistributed and stay under the parent's name
`Nochild.data.aux`	QA	Same, for UMOS

Why the column counts differ between sheets: `Raw.main` / `Raw.aux` have parents and children; `Out.*` have no redistributed parents (only children, plus the no-child parents kept as they are).

3. Column (taxon) names

Names are the `Taxon_Analysis` values from the taxonomy lookup (`Look-up Table_Invert Taxonomy_<date>.xlsx`).

- `Genus.species`: species level (for example `Lymnaea.stagnalis`).
- `Genus.sp.`: genus level, not identified to species.
- `<Family or higher>.UID`: unidentified at that level (for example `Chironomidae.UID`, `Hemiptera.UID`).
- Names such as `Oligochaeta`, `Amphipoda`, `Hydrachnidia`, `Hirudinida` are analysis groups with no finer resolution.
- Raw names that were not in the lookup were mapped to a lookup name with the taxon crosswalk (section 5), so a column can hold specimens recorded under several raw names.

Ambiguous parents (lookup `Analysis_Grouping` = `Distribute.to.Children`) disappear from `Out.*` unless they have no children; the numbers of columns are in section 4.

4. Quick facts per year

Counts are totals over all samples. "UMOS" = unique mature organisms search (a separate search for mature specimens; its counts are not scaled).

	2023	2024	2025
Samples (rows)	95	100	100
Taxa in `Out.*`	77	92	86
Ambiguous parents in data (`AP.info` rows)	42	45	40

	2023	2024	2025
of which no children (kept as is)	17	12	11
Fine (main) total, scaled	671,078	891,607	280,815
UMOS total	398	520	554
`Out.combined` total (APTC)	671,476	892,127	281,369
Most abundant taxon	`Chironomidae.UID` (186,349)	`Chironomidae.UID` (428,699)	`Chironomidae.UID` (105,224)

5. How the numbers were produced

Scripts (run from the project root, in this order): `1_code/r_scripts/01_generic clean WIP data.R` (2025: `01b_clean 2025 WIP data.R`), then `02_build taxa crosswalk.R` if there are new names, then `03_generic resolve WIP data.R`.

1. **Clean**: read the fine, UMOS and Marchant-count tables; stop on missing or duplicated sample IDs.
2. **Name crosswalk**: names not in the lookup are mapped using `1_code/reference_tables/taxa_crosswalk_11.05.2026.csv` (columns: `raw_name`, `lookup_name`, `confidence` high / medium / low, `status` provisional / confirmed). A blank `lookup_name` means the specimens are left out (for example `Terrestrial`, row totals).
3. **Aggregate** raw names to `Taxon_Analysis` and drop `Do.not.analyze`.
4. **Scale fine counts to 100 Marchant cells**: `round(count / cells counted x 100)`. UMOS counts are not scaled.
5. **Resolve** parents with the algorithm in `Taxonomic ambiguous assignment abmi_UMOSfix.R` (a draft fix of the original, see section 7).
6. Export one workbook per run.

6. How to read and use the data

- **Values are standardised counts, not raw counts and not densities.** Fine counts were scaled to a 100-cell Marchant subsample and rounded. A sample where only 5 cells were counted is multiplied by 20, so rare taxa in such samples are inflated.
- Values should be whole numbers. **Totals match the input exactly for APTC. For DPAC, rounding makes totals differ slightly from the input** (2023: +9 specimens over 95 samples, at most 1 per sample; 2024: +2; 2025: 0). See `2_pipeline/WIP <year> resolution mass balance.csv`.
- Use `Out.combined` for community analyses. It includes UMOS specimens.
- Taxa listed in `Nochild.data` (for example `Chironomidae.UID`, `Enallagma.sp.` in 2025) are parents with no children in the lookup, so their counts stay at the parent level. `Chironomidae.UID`

is the largest part of the total in every year (about 28% in 2023, 48% in 2024, 37% in 2025), so be careful with totals and richness that include it.

- **Choice of resolution method changes the numbers.** In 2025, compared with `aptc_abmi_frequent`, `aptc_abmi_abundant` differs in 70 cells of `Out.combined`, `dpac_abmi_frequent` in 98 and `dpac_abmi_abundant` in 160 (of 8,600). Parents with a single child send 100% to that child under every run. Use the same run for all years.
- Samples are not pooled: A/B replicates (2023) are separate rows.